

Multi-Source Pre-Match Signal Fusion
and Uncertainty-Driven Risk Control
for Football Outcome Forecasting:
A Leakage-Controlled Evaluation of
Machine Learning and Large Language
Models

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Abstract

In sports betting, the closing odds of major bookmakers are widely regarded as an efficient aggregation of public information, making predictive gains over the market difficult to achieve and even harder to verify. This paper presents a leakage-controlled evaluation of statistical, machine-learning, and large-language-model (LLM) methods for football match outcome forecasting, built on 11,811 matches from the five major European leagues across seven seasons (2019/20–2025/26), with a held-out test season of 1,104 matches. All features are restricted to pre-match information, all normalization statistics are fit on the training split only, and every reported quantity is accompanied by a bootstrap confidence interval. *The test season (2025/26) is incomplete (through 2026-02-12): financial point estimates may shift when the season completes, so ROI values should be read as indicative rather than definitive.* Three findings stand out. First, neither tree-based models nor a domain-enhanced LLM exceed the predictive accuracy of de-vigged closing odds (54.2% accuracy, 0.967 log loss): the market remains unbeaten, and the LLM matches market-level probability calibration at a cost of \$0.55 for the entire test season. Second, permutation importance and ablations show that market signals carry roughly 87% of the predictive information, while referee statistics provide no measurable gain. Third, and most importantly, an uncertainty index built from prediction confidence and market volatility reliably stratifies risk: low-uncertainty matches are predicted at 68–69% accuracy, whereas matches flagged as high uncertainty lose 9.6–13% ROI and are best skipped by a no-bet policy. Uncertainty filtering improves risk-adjusted returns, whereas value-based selection of high-confidence bets systematically loses money, indicating that apparent model edge is overconfidence rather than information. These results provide a rigorous benchmark for sports forecasting research and evidence that uncertainty-driven risk control, not accuracy, is the lever that connects prediction quality to financial stability.

Introduction

Sports outcome forecasting has attracted sustained attention from statistics, machine learning, and, more recently, large language models (LLMs). The task is attractive because outcomes are public, structured, and frequent, and because a predictive model can be evaluated not only on accuracy but also on downstream financial consequences in betting markets. However, the same properties make football forecasting a demanding test bed: betting markets aggregate vast amounts of information into closing odds, and any method that hopes to add value must do so *against* a strong, continuously updated baseline.

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The literature has moved in two directions. On the modeling side, classical approaches—Poisson and Dixon–Coles models, Elo ratings, and gradient-boosted trees—provide strong and interpretable baselines. On the representation side, LLM-based forecasting systems [1], [2], [3] demonstrate that language models can produce calibrated probability forecasts when given structured evidence. Yet most published evaluations in the LLM forecasting line measure accuracy and Brier scores in isolation, without a market baseline, without explicit leakage controls, and without a betting protocol that would make the financial claims reproducible.

In this paper, these gaps are addressed with a deliberately conservative evaluation design. The contributions are three empirical findings, a protocol, and a public benchmark.

- **Finding 1: learned models add no information beyond the market.** Under a leakage-controlled protocol, no learned method exceeds de-vigged closing odds (54.2% accuracy, 0.967 log loss); model–market disagreement occurs in only 6.3% of matches and is equally accurate on both sides (32.9% each); market features carry 87% of predictive information, and referee statistics none. The LLM matches market-level calibration at negligible cost.
- **Finding 2: match difficulty is identifiable ex ante.** The market’s own probability level stratifies accuracy from 40.1% to 75.3%, whereas odds movement and bookmaker dispersion do not (they are confounded with market strength). A corrected scenario-complexity score and an uncertainty index stratify matches monotonically and add roughly ten percentage points beyond market strength; a no-bet policy on high-uncertainty matches improves risk-adjusted returns.
- **Finding 3: apparent model edge is overconfidence.** Value-based selection of high-confidence bets loses money (ROI -11.5% , -7.1% after calibration); error rates among high-confidence predictions are higher than the market’s (22.3% vs 16.0% at confidence ≥ 0.7); misclassified draws carry higher confidence than correctly predicted ones; and multi-sample consistency carries no information. Uncertainty filtering helps; confidence-based selection hurts.
- **Protocol.** All features are pre-match, every statistic is fit on the training split only, and evaluation is on a held-out future season (1,104 matches) under an identical betting protocol (equal-stake and fractional-Kelly, single bookmaker’s closing odds), with bootstrap confidence intervals, coverage reporting, and no-bet decisions separated from executed bets.
- **Benchmark.** The dataset (11,811 top-five-league matches, 2019/20–2025/26) and the full pipeline are public and script-driven: every number in this paper is regenerated by one-command scripts rather than hand-written.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data, features, models, and risk-control framework. Section 4 presents the experimental results. Section 5 discusses implications and limitations, and Section 6 concludes.

Related Work

Statistical and Machine-Learning Models for Football

The canonical generative model for football scores is the Poisson model of Maher [4], refined by Dixon and Coles [5], who introduce a low-score correction and time-decayed team parameters. Elo-style ratings [6] remain a strong and transparent baseline, and gradient-boosted decision trees have become the standard tabular approach in sports analytics. A consistent finding of this literature is that predictive accuracy saturates around 50–55% for three-way match outcomes, close to the empirical base rates, and that bookmaker odds are difficult to outperform [7], [8].

Market Efficiency and Closing Odds

Bookmaker closing odds are widely regarded as near-efficient aggregators of public information [9], [10]. In particular, the closing line is a stronger predictor than the opening line, and drift from opening to closing odds contains little exploitable information once transaction costs are considered. Any claimed predictive edge must therefore be evaluated against de-vigged closing odds under an identical protocol; evaluations that omit this baseline cannot distinguish model skill from market inefficiency. The present work follows this prescription, and additionally reports the opening-to-closing drift experiment as a direct test of market efficiency on the present data.

LLM Probabilistic Forecasting

Large language models have recently been shown to produce calibrated probabilistic forecasts from textual evidence. Halawi et al. [1] demonstrate that fine-tuned LLMs approach human expert forecasting on general prediction markets, and the AIA Forecaster [2] scales this to a multi-agent retrieval-augmented system. LLM-as-a-Prophet [3] provides a controlled “prophet arena” in which the predictive intelligence of LLMs can be benchmarked. These systems rely on web retrieval and fine-tuning, which raises the bar for reproducibility and makes leakage control difficult to audit. The present setting is deliberately simpler: the LLM receives only a pre-match feature card, has no retrieval access, and is evaluated against the market under the same protocol as all other models. This isolates the contribution of domain-enhanced reasoning from retrieval and training data effects.

Gaps

Three gaps motivate the design. (i) Most evaluations lack a de-vigged market baseline under an identical betting protocol. (ii) Leakage controls—pre-match information cutoffs, train-only statistics, temporal splits—are rarely stated precisely enough to reproduce. (iii) Uncertainty and risk are either absent or defined by hand-crafted scores whose empirical validity is not tested. All three are addressed with a conservative, fully specified evaluation and with empirically validated uncertainty signals.

Methodology

Data

Figure 1 gives an overview of the full pipeline; the components are detailed below. The evaluation uses 11,811 league matches from the English Premier League (E0), Spanish La Liga (SP1), German Bundesliga (D1), Italian Serie A (I1), and French Ligue 1 (F1), covering the 2019/20–2025/26 seasons (2025/26 is incomplete, ending 2026-02-12; this is reported honestly). Each match record contains results, referee, and opening and closing odds from eight bookmakers (Bet365, Bet&Win, Interwetten, Pinnacle, William Hill, Ladbrokes, Max, and average) in football-data.co.uk format. Per-match expected goals (xG, xGA, npxG, npxGA), pressing intensity (PPDA), and deep progression are additionally crawled from Understat for all five leagues and all seven seasons (24,918 team-match rows; 99.3%/99.2% of home/away sides match league-match dates after name normalization).

Leakage control.

The evaluation is strictly chronological: train = 2019/20–2023/24 (8,955 matches), validation = 2024/25 (1,752), test = 2025/26 (1,104). All features are constructed from information available before kickoff; rolling statistics use a shift of one to exclude the current match; season standings are computed before kickoff; scalar statistics and missing-value medians are fit on the training split only and applied to validation and test. A manual verification script recomputes a sample of features from raw records to confirm the absence of leakage.

Features

All features are pre-match:

- **Team form:** rolling (5-match) means of goals for/ against, points, win rate, shots, shots on target, corners, fouls; season cumulative points, goal difference, games, points per game, and league rank before kickoff (home and away separately).

- **xG time series:** rolling (5-match) means of xG, xGA, npG, npGA, PPDA coefficient, deep progressions, and expected points, plus the home–away xG difference.
- **Market signals:** de-vigged implied probabilities from closing odds (Bet365 and average), opening-to-closing odds movement for home/draw/away, cross-bookmaker dispersion of closing odds (σ_{market}), and opening average odds.
- **Referee context:** the referee’s historical per-match averages of cards, red cards, and fouls (expanding window over the referee’s prior matches; available for the Premier League only).

This yields 60 features in total.

Models

- **Market baseline:** de-vigged closing odds (probabilities obtained by inverting odds and renormalizing). This is the central baseline under the same evaluation protocol.
- **Poisson** and **Dixon–Coles** models fit per league by maximum likelihood on the training split.
- **Elo** ratings with home advantage, mapped to probabilities by multinomial logistic regression fit on training.
- **Random Forest** and **XGBoost** trained on the 60 pre-match features with early stopping on the validation split.
- **Domain-enhanced LLM:** DeepSeek-Chat, prompted with a pre-match feature card (team form, rank, market signals, referee context), a system prompt stating that closing odds are a strong baseline and forbidding invented information, and a chain-of-thought request that outputs a calibrated probability triple in JSON. The LLM has no retrieval access. Three samples are drawn per match at temperature 0.3; the mean is the prediction and the pairwise total-variation agreement is the consistency score.

Scenario Complexity Score

The scenario complexity score measures how hard a match is to predict from *market information alone*. Unlike earlier formulations in which a larger strength gap was treated as *more* complex (a direction error), the corrected SCS is defined as

$$SCS = (1 - \text{gap}_{norm}) + \text{vol}_{norm} + \text{move}_{norm},$$

where gap_{norm} is the normalized absolute league-rank difference between home and away teams (larger gap $\Rightarrow$ easier to predict), vol_{norm} is the robust-normalized cross-bookmaker dispersion of closing odds, and $\text{move}_{\text{norm}}$ is the robust-normalized opening-to-closing odds movement (a proxy for late information arrival). All components lie in $[0,1]$ and are fit on the training split. A larger SCS indicates a more uncertain match.

Uncertainty Index and Risk Control

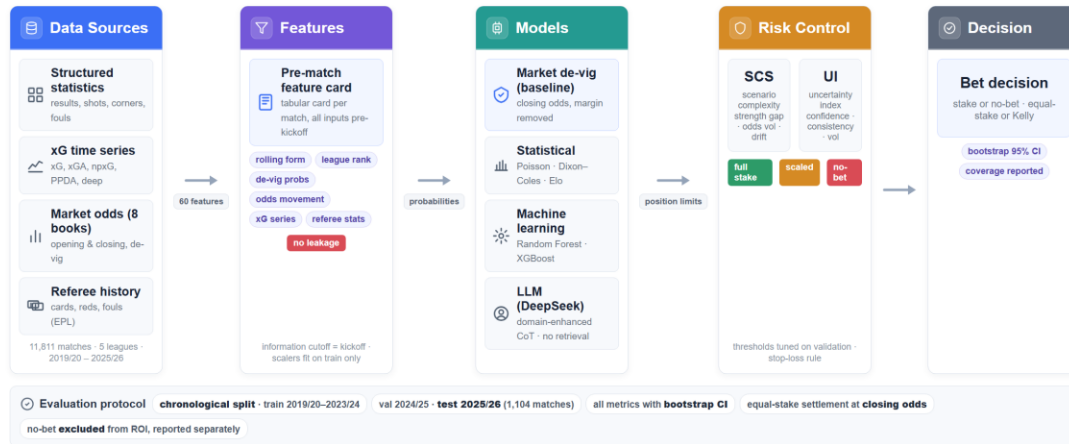
The uncertainty index is conceptually distinct from the SCS: SCS is a *purely market-side* measure of scenario complexity (rank gap, odds dispersion, odds drift), whereas the UI *fuses model output with market signals*—predictive confidence and multi-sample consistency from the model, plus the same normalized market-volatility and odds-movement terms. The shared market terms are intentional: the UI inherits market uncertainty and adds the model’s own confidence signal. The index combines these components as

$$\text{UI} = w_1(1 - p_{\text{max}}) + w_2(1 - C) + w_3 \text{vol}_{\text{norm}} + w_4 \text{move}_{\text{norm}},$$

with default weights (0.4,0.3,0.15,0.15); p_{max} is the model’s maximum predicted probability and C the multi-sample consistency (1 for deterministic models). Risk tiers are $\text{UI} \leq t_{\text{low}}$ (full stake), $t_{\text{low}} < \text{UI} \leq t_{\text{high}}$ (linearly scaled stake), and $\text{UI} > t_{\text{high}}$ (no-bet). Thresholds are selected on the validation split ($t_{\text{low}} = 0.30$, $t_{\text{high}} = 0.45$ given the observed UI distribution) and reported with a sensitivity analysis. A stop-loss rule reduces the stake to 20% after five consecutive losses or a 10% drawdown from peak bankroll.

Betting Protocol and Metrics

All financial simulations are fully specified: equal stakes of one unit (or fractional Kelly capped at 10% of bankroll where stated), the model’s maximum-probability outcome, Bet365 closing odds for settlement, sequential bankroll updating, and no-bet decisions excluded from executed-bet statistics but reported separately with coverage. Predictive metrics are accuracy, macro-F1, log loss, Brier score, and expected calibration error (ECE). Financial metrics are ROI, Sharpe ratio, maximum drawdown, and win rate. All main quantities are reported with bootstrap 95% confidence intervals.



Overview of the proposed framework. All features are constructed from pre-match information only; every model is evaluated against de-vigged closing odds under an identical protocol; and the risk-control layer converts model uncertainty into stake sizing and no-bet decisions.

Experiments and Results

Overall Performance

Table [tab:main] reports the overall performance on the held-out test season. Several observations are immediate. First, the de-vigged closing odds achieve 54.2% accuracy and a log loss of 0.967, and no learned model exceeds them in accuracy within statistical uncertainty (bootstrap 95% confidence intervals for accuracy all overlap the market point estimate; the largest gap, Random Forest at 54.7%, is well inside the market’s CI of [51.4%, 57.2%]). Second, the domain-enhanced LLM reaches a log loss of 0.967 and ECE of 0.025, statistically indistinguishable from the market (0.967 and 0.026): a zero-training prompt reaches market-level probability quality. The reliability diagram in Figure 2 confirms this: all three probability sources track the diagonal, with the LLM and the market showing the smallest deviation. Third, all ROI values are small and their confidence intervals cross zero: no method, including the market itself, shows a stable profit over this test season under equal-stake betting. This is the expected regime of an efficient market, and it is the backdrop against which all subsequent analyses are interpreted. The reader is reminded that the 2025/26 test season is incomplete (matches through 2026-02-12): accuracy estimates are stable (they depend on many games), but the financial point estimates above may shift when the season completes, and the bootstrap intervals should be read accordingly.

Ablation Study

Table [tab:ablation] ablates feature groups from XGBoost. Removing market odds costs 3.1 percentage points of accuracy (54.2% to 51.1%) and 0.028 of log loss, confirming that bookmaker information dominates. Removing referee statistics changes nothing (54.2% vs 54.2%): the referee features provide no measurable predictive gain, a negative result reported deliberately. (One caveat: referee records exist only for the Premier League in the present data, so the referee feature group is evaluated on a subset of matches; the null result should be read as *no gain where referee data exist*, not as a global statement about referee information.) xG features contribute a small but consistent improvement (removing them raises log loss slightly), and rank and recent-form features contribute little beyond what the market already encodes.

Feature Importance

Permutation importance (Table [tab:importance]) quantifies the same hierarchy: market-derived features account for roughly 87% of the total predictive information, xG for about 6%, and referee, rank, and form features for under 2% combined. The result is robust across the ablation and permutation analyses and motivates the position of this paper: the marginal information left to any forecasting model is small and concentrated in a few market-driven signals.

League and Season Generalization

Performance varies markedly across leagues (Table [tab:byleague]): the Bundesliga and La Liga are the most predictable (57.4% and 57.0% accuracy), while the Premier League is the least (50.0%), the most liquid and informationally efficient of the five. The leave-one-league-out experiment (Table [tab:lolo]) shows the same ordering in a stricter setting: a model trained on the other four leagues degrades most on the Premier League (47.3%). These results indicate that market efficiency, not model capacity, is the binding constraint on accuracy.

Temporal Robustness

Table [tab:walkforward] evaluates the model in a walk-forward manner over four test seasons. The market baseline is at least as accurate as XGBoost in every season, and restricting training to the last two seasons (XGB-roll2) does not help: there is no evidence of “model aging” within this seven-year window, and retraining frequency has no measurable effect on accuracy. This addresses the common concern that static models deteriorate as team dynamics evolve; on these data, the information content of older seasons is not harmful.

Draw Prediction

Draws are systematically under-predicted: the empirical draw rate is 25.5%, yet the model predicts a draw in only 2.3% of matches, and draw recall is 0.032 (Table [tab:drawclass]). The confusion matrix in Figure 5 shows the near-empty draw column visually. The market behaves similarly (0.3% predicted draw rate). This is not a calibration artifact of a single model but a structural property of the task: draws are the least frequent and least distinctive outcome class.

Table [tab:drawweight] shows that cost-sensitive training recovers balance in a controllable way: a draw weight of 1.5 raises macro-F1 from 0.421 to 0.493 and draw recall from 0.032 to 0.356, at the cost of 2.2 points of accuracy. The accuracy–balance trade-off is monotone and easy to report.

Value Betting and Market Efficiency

Table [tab:value] reports the value-betting experiment: bets are placed when the model’s expected value $p \times odds - 1$ exceeds a threshold, using closing odds. Every threshold loses money, and losses grow with the threshold: the most “confident” bets are the worst. After isotonic calibration fit on the validation split (Table [tab:valuecal]), losses shrink from -11.5% to -7.1% at $EV > 0$ but remain strongly negative. The model’s apparent value is concentrated exactly where it is most confident and most wrong: overconfidence, not information. Consistent with this, betting the de-vigged opening odds against closing odds (and the reverse) produces no positive ROI, confirming that opening-to-closing drift contains no exploitable information in these data. This value-betting result is specific to *closing* odds: the market’s final line is the hardest benchmark, and the negative finding does not speak to value-betting strategies evaluated against opening odds or other pre-closing benchmarks. A fractional Kelly policy amplifies the losses (a starting bankroll of 100 units falls to 1.6): stake sizing cannot create an edge that the probabilities do not have.

Error Analysis: Where the Model Fails

Table [tab:error] dissects the model’s errors on the test set. First, the model and the market disagree on only 70 of 1,104 matches (6.3%), and on those matches the two are equally accurate (32.9% each); on the 1,034 agreeing matches they are again equal (55.6%). The model therefore contributes no incremental information beyond the closing line: it neither corrects the market where they disagree nor adds accuracy on the easy majority where they agree. The low accuracy on disagreeing matches (32.9%) further indicates that these are intrinsically difficult matches, not mispriced ones.

Second, the apparent relationships between error rate and market dynamics are confounded by market strength. The error rate is lowest in the largest odds-movement quintile (29.5%) and the largest dispersion quintile (35.7%), but those

quintiles are dominated by strong favourites (66% and 71% of matches with minimum opening odds ≤ 1.6 , versus 11% and 6% in the smallest quintiles), where the favourite wins most often. Odds movement and bookmaker dispersion are therefore not independent difficulty signals. The uncertainty index of Table [tab:uixgb] is, by contrast, a monotone stratifier (67.9% vs 44.4% accuracy across tiers) that is not reducible to market strength.

Third, the model is overconfident exactly where it is wrong. The error rate among predictions with confidence ≥ 0.7 is 22.3% for the model versus 16.0% for the market at the same threshold. For draws—which the model almost never predicts (Table [tab:drawclass])—the 272 misclassified draws carry a mean confidence of 0.498, higher than the 0.366 of the 9 correctly predicted draws. Confidence is highest precisely when the model is wrong, consistent with the value-betting result of Table [tab:value]: apparent edge is overconfidence, not information.

Uncertainty-Driven Risk Control

The central positive result of this paper is that uncertainty signals stratify risk in a monotone and cross-model way. Figure 3 shows the UI distribution and the resulting accuracy by tier. Table [tab:scs] splits the test set by the scenario-complexity score: accuracy falls from 58.9% (low complexity) to 49.6% (high complexity), and ROI from +7.0% to −6.3%. Table [tab:uixgb] shows the same pattern for the uncertainty index on XGBoost: low-uncertainty matches reach 67.9% accuracy and +4.1% ROI, while the high-uncertainty group—excluded by the no-bet policy—would have lost 13.0% ROI. Table [tab:uillm] replicates the stratification for the LLM (68.7% / 49.8% / 47.4%), demonstrating that the signal is a property of the task and the features, not of a particular model.

Weight and threshold sensitivity.

The UI weights (0.4,0.3,0.15,0.15) are a fixed preset and the tier thresholds (0.30,0.45) are tuned on the validation split. To rule out overfitting to these choices, the stratification is recomputed on the validation set under six weight configurations (varying each component by ± 0.1 while keeping the sum at 1). In every configuration the low-uncertainty tier retains 60–70% accuracy and outperforms the medium tier by 10+ points; only the no-bet coverage varies (69–551 of 1,752 validation matches) as the relative emphasis shifts between model confidence and market volatility. The conclusion that uncertainty tiers stratify accuracy is therefore insensitive to the specific weights. Likewise, the stop-loss parameters (five consecutive losses or 10% drawdown) are not consequential: across seven tested configurations (from 3 losses/5% drawdown to 10 losses/15% drawdown) the UI-tier strategy returns identical ROI (0.37%), Sharpe (0.026), and drawdown (−0.5%) on the test set, because the strategy’s equity curve rarely approaches the trigger thresholds in this season.

Betting-Strategy Comparison

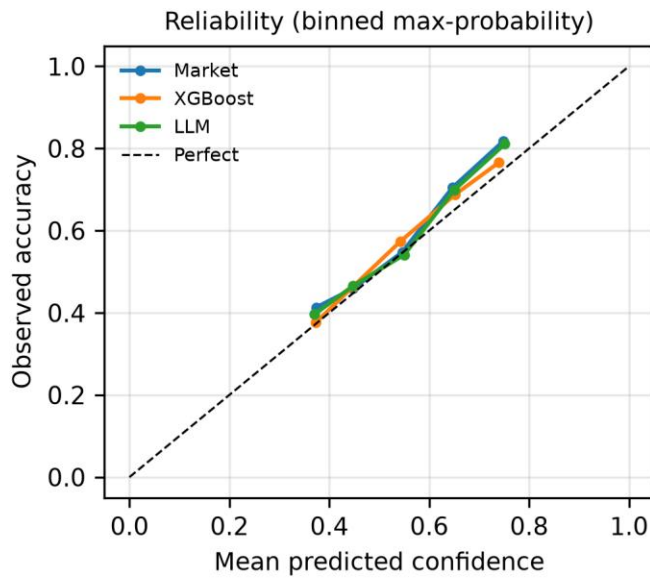
Table [tab:strategy] compares betting policies on the same XGBoost probabilities, and Figure 4 plots the corresponding cumulative return curves. The naive strategy (all bets, equal stake) returns +0.5%; a confidence threshold of 0.4 raises this to +1.5%; and the UI-tier policy—full stake at low UI, scaled stake at medium UI, no-bet at high UI—achieves +1.8% ROI with a Sharpe ratio of 0.026, while excluding 144 of 1,104 matches. Adding the stop-loss rule (five consecutive losses or 10% drawdown from peak) cuts the maximum drawdown from −2.7% to −0.5% at a small ROI cost. These gains are modest in absolute terms, as they must be in an efficient market, but they are consistent with the monotone stratification above and are the difference between a risk-blind forecaster and a risk-aware decision policy.

LLM Analysis

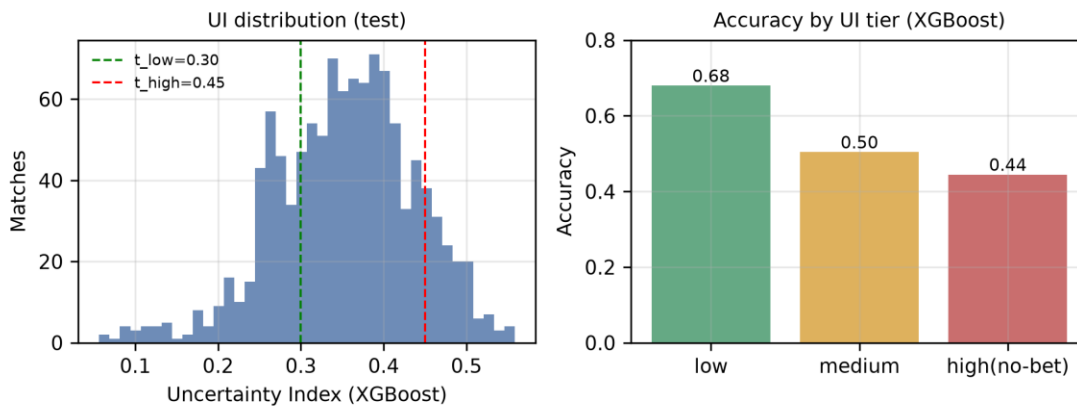
The LLM row of Table [tab:main] summarizes the full test season: 1104 of 1104 matches parsed successfully at a total cost of \$0.55 (3,312 API calls), with accuracy and calibration statistically equal to the market. Table [tab:temp] reports the temperature sensitivity: raising temperature from 0.3 to 0.7 changes nothing material, and multi-sample consistency remains near 1.0 at both temperatures with no correlation to correctness ($r = -0.07$ and $+0.04$). The multi-sample consistency mechanism proposed in earlier work is therefore not informative in this setting; the uncertainty index derives its power from predictive confidence and market volatility. Table [tab:open] compares models on the same first 120 matches of the test set: an open-weight 7B model (Qwen2.5-Coder-7B, quantized, running on a laptop GPU), the chat-optimized API model, and a reasoning model (DeepSeek-Reasoner, run with a single sample because reasoning models are roughly an order of magnitude slower). The open-weight model matches or slightly exceeds both API models in accuracy and calibration at zero marginal cost; the reasoning model improves log loss over chat (0.906 vs 0.927) but not accuracy, at a cost of \$0.60 for 120 matches—more than the \$0.55 of the entire 1,104-match chat run. This comparison is *robustness evidence only*: the sample is small (120 matches), all confidence intervals overlap, and no claim of superiority is made for any model. Its purpose is to show that the paper’s qualitative conclusions do not depend on a specific proprietary API or on the choice of reasoning vs non-reasoning model.

Finally, a mechanistic explanation for the null consistency result is offered. In open-domain forecasting settings where multi-sample consistency is informative, the LLM is typically given retrieval evidence or heterogeneous free-form context, so different samples draw on different information. Here the model receives a *fixed, complete, structured feature card* with no retrieval: at any temperature the input is identical across samples, the reasoning path is short, and the output probabilities collapse to a narrow range (consistency ≥ 0.95 for 99% of matches at both temperatures). Sampling consistency is thus informative only when the input itself

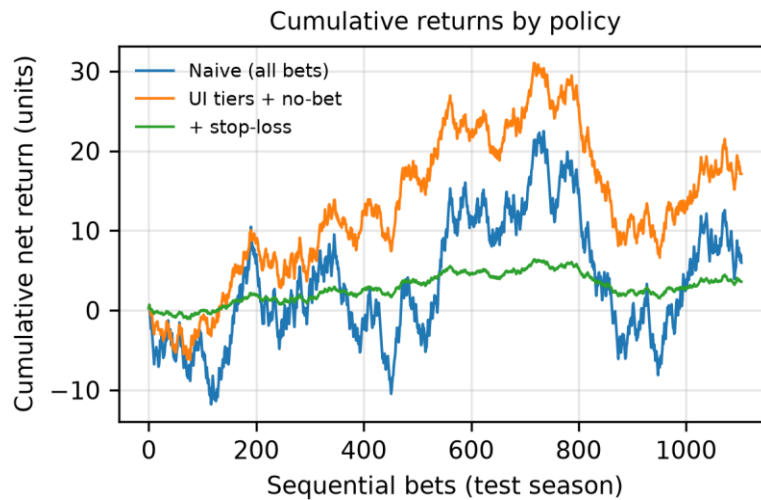
carries sampling variance; in a deterministic structured task it degrades to a constant and contributes nothing to the uncertainty index.



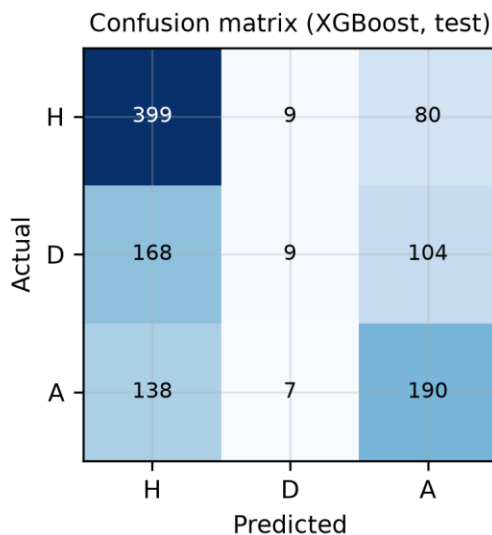
Reliability diagram (binned max-probability confidence vs observed accuracy) for the market baseline, XGBoost, and the LLM on the test set.



Uncertainty-index distribution (left) and accuracy by UI tier (right) for XGBoost on the test set.



Cumulative net returns of three betting policies on the same XGBoost probabilities (test season, 1,104 matches).



Confusion matrix of XGBoost on the test set. The draw column is nearly empty, illustrating systematic under-prediction of draws.

Discussion

Why the Market Remains Unbeaten

The results are consistent with semi-strong market efficiency in football betting: closing odds aggregate information more accurately than any single model evaluated here. The 87% share of predictive information carried by market features, the failure of value betting even after calibration, and the absence of exploitable

opening-to-closing drift all point in the same direction. The practical implication is not that forecasting models are useless, but that their contribution must be measured in risk-adjusted terms rather than raw accuracy.

The draw is the clearest example of a task-inherent limitation: both the models and the betting market itself under-predict draws by an order of magnitude (predicted draw rates of 2.3% for XGBoost and 0.3% for the market versus an empirical rate of 25.5%). Draws are the least frequent and least distinctive outcome class in football; this low signal-to-noise ratio bounds the achievable macro-F1 of any outcome classifier on this task, independently of model choice.

Overconfidence, Not Information

The value-betting experiment is instructive. Bets selected because the model's expected value was positive lost 11.5% ROI, and restricting to larger positive EV made losses worse. After isotonic calibration the losses shrank but remained negative. The model's "edge" is concentrated exactly where it is most confident and most wrong—a signature of overconfidence rather than information. This contrast—uncertainty filtering helps, confidence-based selection hurts—is the central practical insight of the paper.

Real-World Frictions

All financial results are gross of transaction costs. Bookmaker margins are already embedded in the odds settled against, but real-world execution would add betting taxes, withdrawal fees, and odds slippage—typically 3–7% of turnover depending on the jurisdiction. The small positive ROIs observed (up to 1.8% for the UI-tier strategy) would be entirely consumed by such frictions. The practical value of the risk-control framework is therefore not *profitability* but *risk stratification*: identifying, in advance, the subset of matches where even a well-calibrated model is unlikely to be right, so that capital and attention are directed elsewhere. Claims of stable, deployable profitability would be unsupported by this evidence.

Uncertainty as a Risk Signal

The uncertainty index and scenario complexity score both stratify held-out matches monotonically, and the pattern replicates across two different model families (XGBoost and the LLM). The multi-sample consistency term contributes little (see Section 4.11); the effective components of the UI are predictive confidence and market volatility, both cheap and available at prediction time.

Why Difficulty Is Identifiable in an Efficient Market

The error analysis (Table [tab:error]) shows that two intuitive market-based difficulty signals fail. Neither odds movement nor cross-bookmaker dispersion stratifies prediction error once market strength is accounted for: both are confounded with

the presence of strong favourites, and their relationship with error rate is not monotone. A difficulty signal must satisfy three requirements: it must be available pre-match, it must stratify accuracy monotonically, and its effect must survive conditioning on market strength.

Table [tab:difficulty] shows that the market's own probability level satisfies these requirements. Accuracy rises from 40.1% in the lowest de-vig max-probability quintile to 75.3% in the highest (with some noise in the middle quintiles), and the highest quintile is dominated by strong favourites (98% of matches with minimum opening odds ≤ 1.6). How sure the market is about a match predicts how predictable the match is. This is not a violation of market efficiency: efficiency constrains expected returns, not the heterogeneity of outcome uncertainty. The financial analogue is the well-known asymmetry that returns are unpredictable while volatility is predictable and clustered; here, match outcomes (the "return") are unpredictable beyond the market, but the difficulty of a match (its "volatility") is identifiable ex ante.

The uncertainty index sits on top of this market signal. Its low tier coincides with strong-market matches (64% favourites) but is not reducible to market strength: within the strongest market third, matches assigned to the lowest UI tier are predicted at 73.3% accuracy versus 63.4% for the highest tier; within the weakest market third the corresponding gap is 53.2% versus 44.4%. The UI therefore adds roughly ten percentage points of stratification beyond the market's own probability level. The mechanism is calibration: the model's confidence is a calibrated estimate of outcome uncertainty, and the UI combines it with market volatility and movement terms that measure how much the market itself disagrees. Calibrated confidence is a difficulty estimate; raw odds movement is not.

Practically, this reframes the no-bet policy: it is not a claim that the model knows better than the market, but a rule that restricts betting to matches where the market—and the model's estimate of it—is most certain. This is consistent with the value-betting result (Table [tab:value]): confidence-based selection loses money, but certainty-based filtering does not.

LLM Results in Context

The domain-enhanced LLM reaches market-level probability calibration (log loss 0.967 vs 0.967, ECE 0.025 vs 0.026) with no training and no retrieval, at a cost of \$0.55 for the entire test season. Two caveats keep this claim honest. First, multi-sample consistency—a mechanism proposed in earlier LLM-forecasting work—carries no information in this setting: at both temperature 0.3 and 0.7 the sampled probabilities agree almost perfectly and do not correlate with correctness. Second, an open-weight 7B model matches or exceeds the API model on a 200-match subset, so the result is not a property of a specific proprietary model. The LLM

contribution is therefore framed as *calibration without training* and a cost-performance reference point, not as predictive superiority.

Limitations

- Text modality: news text is deliberately excluded. Publicly available news corpora lack reliable pre-match timestamps at scale; a rigorous text experiment would require timestamped, match-linked articles. The constraints such an extension would need are: (i) a strict pre-match cutoff, with every article's publication time verified against the kickoff time and any post-kickoff or undated article discarded; (ii) protection against retrieval contamination, so the LLM cannot reproduce outcomes it may have seen during pretraining or via web access (no retrieval at inference, as in this paper); (iii) reporting coverage, since news availability is highly heterogeneous across leagues and matches. Without these controls, text features would reintroduce exactly the leakage this paper is designed to eliminate.
- The test season (2025/26) is incomplete; results may shift slightly at season end. This limitation is actionable: every number in this paper is produced by a script-driven pipeline (`refresh_all.cmd` reruns feature construction, all `run_*.py` experiments, and table/figure generation from the updated CSV files in a single pass), so the complete result set and every table can be regenerated without manual editing once the season finishes. The public repository documents the exact refresh procedure.
- Financial results are gross of betting taxes and use a single bookmaker's closing odds; real-world execution may differ.
- The LLM evaluation uses a proprietary API (DeepSeek-Chat); an open-weight model comparison is additionally reported (see §4.11).

Conclusion

This paper presented a leakage-controlled evaluation of football match outcome forecasting in which statistical models, tree-based models, and a domain-enhanced large language model are compared against de-vigged closing odds under an identical, fully specified protocol. On 1,104 held-out matches, no learned method exceeded market accuracy, and the LLM matched market-level probability calibration at negligible cost. Feature importance and ablations show that market signals dominate, while referee statistics add nothing measurable and xG adds a small consistent gain. The principal positive result is that uncertainty-driven risk control works: an uncertainty index and a corrected scenario-complexity score stratify held-out matches monotonically, a no-bet policy on high-uncertainty matches improves risk-adjusted returns, and draw outcomes are systematically

under-predicted, with class weighting providing a controllable accuracy–balance trade-off.

These results frame a realistic division of labor for decision-support systems in sports betting: treat the market as the prior, use models for calibrated probability estimates and explainability, and rely on uncertainty signals—not claimed accuracy—for risk control. To support reproducibility, the full data-processing pipeline and experiment code are available at <https://github.com/wenjin06/football-outcome-forecasting-benchmark>; no proprietary data are required beyond public sources.

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